INCH-POUND

MIL-DTL-16377/77A(SH)
18 December 2008
SUPERSEDING
MIL-DTL-16377/77(SH)
27 September 2004

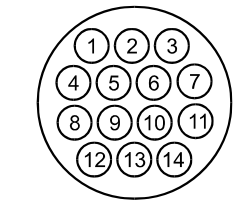
DETAIL SPECIFICATION SHEET

FIXTURES, LIGHTING; EMERGENCY BALLAST, FOR ONE 17- OR 20-WATT T12 OR 17-WATT T8 FLUORESCENT LAMP

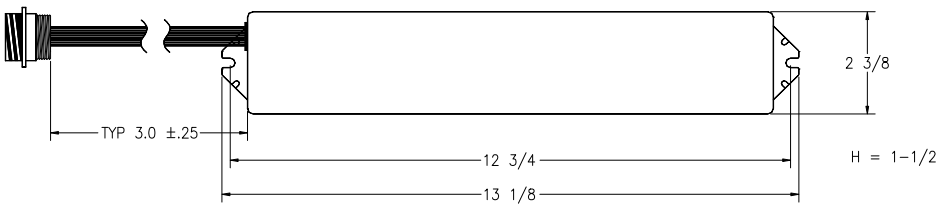
This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-16377.

All dimensions in inches. For tolerances, other than specified herein, see 3.5.2 of MIL-DTL-16377.



CONTACT ARRANGEMENT



CONTACT ARRANGEMENT	WIRE COLOR
1	Red w/ White Stripe
2	White w/ Red Stripe
3	Red w/ White Stripe
4	No Connection
5	White w/ Red Stripe
6	Blue w/ White Stripe
7	No Connection
8	No Connection
9	Blue
10	White
11	No Connection
12	Green
13	Red
14	Yellow

LABEL TYPE I

BALLAST MIL-DTL-16377/77 **NSN 6250-01-491-7039** **120 VOLT, 60 HERTZ** **OPERATES ONE 17- to 20-WATT T-12 OR 17-WATT T8 LAMP** **CONTAINS NICKEL-CADMIUM BATTERY**	**PROPER DISPOSAL REQUIRED** **DATE OF MANUFACTURE** _____ **MANUFACTURER'S IDENTIFICATION OR TRADEMARK**

2"

4"

FIGURE 1. Emergency ballast with wiring harness.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:55Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

DESIGN:

Electrical rating:	17-20 watts, 120 volts, 60 hertz.
Circuit design:	Ballast circuit shall incorporate Thyristor devices for protection against high voltage spikes.
Dimension and configurations:	See figure 1.
Assembly:	The emergency ballast consists of a high temperature, rechargeable NI-CAD battery; battery charger; AC power sensing circuit; solid-state inverter; low voltage battery cutoff, wiring harness, and enclosure. UL approved ballast may be furnished if it conforms to the dimensions and requirements specified.
Weight:	4-pound maximum.
Performance characteristics:	
Input voltage	120 VAC±10 percent.
Emergency operating time	5.0 hours minimum.
Full recharge time	20 hours.
Hi-temp NI-CAD battery	24 watt hours at 55 °C.
Battery rated voltage	6.0 volts.
Battery discharge current	0.8 amps.
Battery charge current	0.2 amps.
Charging component of AC line	0.04 amps.
Charging power	4.0 watts.
System power factor	0.9 minute.
System harmonics	<3 percent.
Low voltage cutoff	4.0 volts.
Lumen output at 0 hours	240 lumens.
Lumen output at 5 hours	144 lumens.

DESIGNATION AND MARKING:

One Type I label, equipment identification, reference MIL-DTL-16377 figure 11, as specified in figure 1.

EXAMINATIONS AND TESTS:

Examinations and tests shall be as specified in MIL-DTL-16377 and the following:

First article testing/inspection:	See table II.
Comparison inspection:	See table II.
Conformance inspection:	See MIL-DTL-16377.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:55Z
Check the source to verify that this is the current version before use.

Inspection of product for delivery:

Examination:	See MIL-DTL-16377.
Conformance test:	See MIL-DTL-16377.
Examination for preparation of delivery:	See MIL-DTL-16377.

GENERAL INFORMATION:

Table I. Standard parts.

Table II. First article tests.

NOTES:

1. Ballast shall conform to marking requirements and color coding of leads as specified in the applicable paragraph of NEMA C82.1.
2. T8 Emergency Ballast Only - Installation circuit wiring shall incorporate a minimum ratio of eight 17-watt lamps and one emergency 17-watt lamp. The 8-to-1 ratio is required for meeting MIL-STD-1399-300 harmonic requirements. The 17-watt emergency ballast produces non-linear current that must be offset by the linear current from eight 17-watt lamp ballasts connected to the same circuit.
3. Emergency fixtures may be designated as Navy Emergency Automatic Lighting System (NEALS).

TABLE I. Standard part.

Name	Military p/n or specification	National Stock Number (NSN)	Remarks
Emergency ballast, wiring harnesses and mating connector, 17- or 20-watt T12 or 17-watt T8 fluorescent lamp	M16377/77-001	6250-01-491-7039	Complete assembly: includes wiring harnesses and matting AMP #206043 connectors.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:55Z
Check the source to verify that this is the current version before use.

TABLE II. First article tests and comparison inspection for ballast.[1/]

Test	Requirement	Test method	Remarks
Examination	3.13	4.5	---
Operation	3.13	4.8.1	---
Electromagnetic interference	3.6.14	4.8.16	---
Electrical equipment interface testing requirements: MIL-STD-1399-300	MIL-STD-1399-300	MIL-STD-1399-300	2/, 3/
Dielectric withstanding voltage	3.6.15	4.8.2	4/
Insulation resistance	3.6.16	4.8.3	4/
Shock	3.5.12	4.8.7	4/
Vibration	3.5.13	4.8.8	4/
Dielectric withstanding voltage	3.6.15	4.8.2	4/
Insulation resistance	3.6.16	4.8.3	4/
Noise	3.5.14	4.8.9	---
Performance a. Power output b. Lumen output c. System power factor d. Operating time	5/	6/	7/
Battery discharge test	8/	8/	---

NOTES:

1/ Requirements and test methods use MIL-DTL-16377.

2/ If less, then 1 KVA exception not applicable for input waveform THD.

3/ Solid state components shall be subject to 250 positive spikes and 250 negative spikes (500 total spikes), one spike per minute, using testing method identified in MIL-STD-1399-300.

4/ Tests shall be performed in the order specified.

5/ The performance of the ballast shall be in accordance with the requirements listed in the applicable paragraph of NEMA C82.2.

6/ Test procedure for performance shall be in accordance with NEMA C82.1.

7/ Sample lot size for inspection shall consist of five ballast units.

8/ Battery discharge test shall be performed for first article testing. Testing shall include the following:

The storage battery shall be capable of supplying and maintaining the maximum total associated load for a period of no less than 5 hours and no less than 87.5 percent rated battery voltage. To determine if the unit complies with this requirement, the battery is to be tested at 55 °C with its associated charging equipment as follows:

a. The battery shall be charged to the maximum rated battery voltage in a period of time to not exceed 24 hours.

b. The charged battery shall then be connected to its maximum rated load and permitted to discharge for 5 hours. At the end of the discharge, the battery voltage shall be no less than 87.5 percent or the rated battery voltage or the illumination no less than 60 percent of initial illumination.

c. At the end of the discharge period, the battery shall be charged for a period of time not to exceed 24 hours.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:55Z
Check the source to verify that this is the current version before use.

TABLE II. First article tests and comparison inspection for ballast. – Continued.

d.	The battery shall be again discharged at maximum rated load for 3.33 hours, and the battery voltage shall be no less the 87.5 percent of the rated battery voltage or the illumination no less than 60 percent of initial illumination at the end of 3.33 hours.
e.	Following the discharge, the battery again shall be recharged to rated battery voltage in a time period not to exceed 24 hours.
f.	The battery then shall be discharged while connected to maximum rated load until “low voltage cutoff” engages. The minimum time for cutoff shall be no less than 5 hours.
g.	At the end of the discharge time, the battery shall be recharged to rated battery voltage in a time period not to exceed 24 hours.
h.	Following recharging, the battery shall be discharged at rated load for 5 hours. At the end of the discharge, the battery voltage shall be no less than 87.5 percent rated battery voltage or the illumination no less than 60 percent of initial illumination.

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue, due to the extent of the changes.

Preparing Activity:
Navy – SH
(Project 6250-2008-001)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:55Z
Check the source to verify that this is the current version before use.